

Displays

D1:

$$\begin{aligned} & \underset{x \in \mathcal{X}}{\text{minimize}} && f(x) \\ & \text{subject to} && x \in \bigcap_{i=1, \dots, N} \mathcal{X}_{\delta^{(i)}} \end{aligned} \quad (1)$$

D2:

$$\epsilon(s_N^*) := \begin{cases} 1 & \text{if } s_N^* = N, \\ 1 - \sqrt[N-s_N^*]{\frac{\beta}{N \binom{N}{s_N^*}}} & \text{otherwise.} \end{cases} \quad (2)$$

D3:

$$\mathbf{P}\{V(x_N^*) > \epsilon(s_N^*)\} \leq \beta. \quad (3)$$

D4:

$$\epsilon(s_N^*) := \begin{cases} 1 & \text{if } s_N^* \geq d, \\ 1 - \sqrt[N-s_N^*]{\frac{\beta}{d \binom{N}{s_N^*}}} & \text{otherwise.} \end{cases} \quad (4)$$

D5:

$$\mathcal{R}(\theta) = \{x \in \mathbb{R}^{n_x} : g(x, \theta) \leq 0\} \quad (5)$$

D6:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\theta) \\ & \text{subject to} && \theta \in \bigcap_{i=1, \dots, N} \{\theta_i : g(\delta^{(i)}, \theta_i) \leq 0\} \end{aligned} \quad (6)$$

D7:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\theta) \\ & \text{subject to} && g(\delta^{(i)}, \theta) \leq 0, i = 1, \dots, N \\ & && \theta \in \mathbb{R}^{n_\theta}. \end{aligned} \quad (7)$$

D8:

$$g(x, \theta) = \sum_{i=1}^m \theta_i f_i(x) \quad (8)$$

D9:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && -\sum_{i=1}^m \theta_i \\ & \text{subject to} && \sum_{i=1}^m \theta_i \mathbf{1}_{A_i}(\delta^{(j)}) \leq 0, \quad j = 1, \dots, N \\ & && \theta \in [0, 1]^m, \end{aligned} \quad (9)$$

D10:

$$f(x, \mu, \sigma) = e^{-\frac{1}{2} \frac{(x - \mu_i)^2}{\sigma_i^2}} \quad (10)$$

D11:

$$g(x, \theta) = \sum_{i=1}^m f(x, \mu_i, \sigma_i) - \gamma \quad (11)$$

D12:

$$\begin{aligned} & \underset{\mu, \sigma}{\text{minimize}} && \sum_{i=1}^m \sigma_i^2 \\ & \text{subject to} && \sum_{i=1}^m e^{-\frac{1}{2} \frac{(\delta^{(j)} - \mu_i)^2}{\sigma_i^2}} - \gamma \geq 0, \quad j = 1, \dots, N, \\ & && \sigma \in [0, \infty)^m \end{aligned} \quad (12)$$

D13:

$$\ddot{x} = u_1 K \sin(\theta), \quad \ddot{h} = -g + u_1 K \cos(\theta), \quad \ddot{\theta} = -d_0 \theta - d_1 \dot{\theta} + n_0 u_2 \quad (13)$$

D14:

$$\begin{aligned} x(0) &\in [-1.7, 1.7], & h(0) &\in [0.3, 2.0], & \theta(0) &\in [-\pi/12, \pi/12], \\ \dot{x}(0) &\in [-0.8, 0.8], & \dot{h}(0) &\in [-1.0, 1.0], & \dot{\theta}(0) &\in [-\pi/2, \pi/2], \end{aligned} \quad (14)$$

Inline

I1: Δ

I2: σ

I3: $\mathbf{P}$

I4: δ

I5: $\delta^{(i)}$

I6: $x \in \mathcal{X} \subseteq \mathbb{R}^d$

I7: $\mathcal{X}_{\delta^{(i)}}$

I8: $\delta^{(1)}, \dots, \delta^{(N)}$

I9: N

I10: f

I11: $\mathcal{X}_\delta$

I12: $x \in \mathcal{X}$

I13: $V(x) = \mathbf{P}\{\delta \in \Delta : x \notin \mathcal{X}_\delta\}$

I14: x_N^*

I15: $V(x_N^*)$

I16: $V(x_N^*) \leq \epsilon$

I17: ϵ

I18: s_N^*

I19: $\epsilon(s_N^*)$

I20: β

I21: $\beta \in (0, 1)$

I22: $s_N^* \in \{0, 1, \dots, N\}$

I23: $s_N^* < d$

I24: d

I25: $\mathcal{R} = \{\Phi(t_1; t_0, x_0, d) : x_0 \in \mathcal{X}_0, d \in \mathcal{D}\}$

I26: $\mathcal{X}_0 \subseteq \mathbb{R}^{n_x}$

I27: $\mathcal{D}$

I28: $d : [t_0, t_1] \rightarrow \mathbb{R}^{n_d}$

I29: $\Phi : \mathcal{X}_0 \times \mathcal{D} \rightarrow \mathbb{R}^{n_x}$

I30: t_1

I31: $\mathcal{X}_0$

I32: t_0

I33: $\mathcal{R}$

I34: $X_0 \in \mathcal{X}_0$

I35: $D \in \mathcal{D}$

I36: $R = \Phi(t_1; t_0, X_0, D)$

I37: $\epsilon \in (0, 1)$

I38: $\delta^{(i)} = \Phi(t_1; t_0, x_{0i}, d_i), i = 1, \dots, N$

I39: $x_{01}, \dots, x_{0N} \stackrel{i.i.d}{\sim} X_0$

I40: $d_1, \dots, d_N \stackrel{i.i.d}{\sim} D$

I41: $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}$

I42: θ

I43: $\text{Vol} : \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}$

I44: $\mathcal{R}(\theta)$

I45: $V(\mathcal{R}(\theta))$

I46: $V(\mathcal{R}(\theta)) > \epsilon$

I47: $\text{Vol}(\theta)$

I48: $\delta^{(1)}, \dots, \delta^{(N)}$
 I49: $\mathbf{P}\{V(\mathcal{R}(\theta)) > \epsilon\} \leq \beta$
 I50: $\epsilon, \beta \in (0, 1)$
 I51: $g(x, \theta)$
 I52: x
 I53: $f_1(x), \dots, f_m(x) : \mathbb{R}^D \rightarrow \mathbb{R}$
 I54: g
 I55: $A \subseteq \mathbb{R}^D$
 I56: m
 I57: $A_1, \dots, A_m$
 I58: $\cup_{i=1}^m A_i = A$
 I59: $A_i \cap A_j = \emptyset \forall i$
 I60: $\mathbf{1}_{A_i}$
 I61: A_i
 I62: $\mathbf{1}_{A_i}(x) = 1$
 I63: $x \in A_i$
 I64: $\mathbf{1}_{A_i}(x) = 0$
 I65: $\theta_i = 0 \forall i$
 I66: $\delta^{(j)} \in A_i$
 I67: $j \in \{1, \dots, N\}$
 I68: $\theta \in [0, 1]^m$
 I69: $\theta_i = 1$
 I70: $\delta^{(j)}$
 I71: $\Phi(t_1; t_0, x_0, d)$
 I72: X_0
 I73: D
 I74: B
 I75: $\theta_1, \dots, \theta_m$
 I76: A_j
 I77: $A_j \in \mathcal{R}(\theta)$
 I78: $\theta = 0$
 I79: $\theta_j = 1$
 I80: $N = (B \cdot \text{number of iterations})$
 I81: $i \in \{1, \dots, N\}$
 I82: $x_{0i} \sim X_0, d_i \sim D$
 I83: $\delta^{(i)} = \phi(t_1; t_0, x_{0i}, d_i)$
 I84: $\delta^{(i)} \in A_i$
 I85: $\theta_i = 0$
 I86: $s_N^* = |\mathcal{R}(\theta)|$
 I87: $d = m$
 I88: $\theta_1, \dots, \theta_m$
 I89: $f(x, \mu, \sigma)$
 I90: $x, \mu, \sigma,$
 I91: μ
 I92: $\theta = (\mu_1, \dots, \mu_m; \sigma_1, \dots, \sigma_m; \gamma)$
 I93: γ
 I94: $\phi(t_1; t_0, x_0, d)$
 I95: Γ
 I96: $\Sigma = \{\sigma_1, \dots, \sigma_m\}$
 I97: $\mu_i, \dots, \mu_m$
 I98: $S = 0$
 I99: $\gamma = \Gamma$
 I100: $\mu_1, \dots, \mu_m =$
 I101: $\sigma_1, \dots, \sigma_m$
 I102: $c = e^{-\frac{1}{2} \frac{(\delta^{(i)} - \mu_i)^2}{\sigma_i^2}} - \gamma$
 I103: $c \geq 0$

I104: $\sigma_i = 0$
 I105: $\sigma_i = \infty$
 I106: $\Sigma = \{\sigma_1, \dots, \sigma_m\}$
 I107: $j \in \{1, \dots, N\} \setminus i$
 I108: $c = e^{-\frac{1}{2} \frac{(\delta^{(j)} - \mu_j)^2}{\sigma_j^2}} - \gamma$
 I109: $\sigma_j = 0$
 I110: $\sigma_j = \infty$
 I111: $\Sigma_i = \{\sigma_1, \dots, \sigma_m\}$
 I112: $\Sigma_i = \Sigma$
 I113: $S = S + 1$
 I114: $s_N^* = S$
 I115: Σ
 I116: 46,052
 I117: .01
 I118: 0.9999
 I119: $\ddot{x} = -\alpha y + x - x^3 + \gamma \cos(\omega t)$
 I120: $x, y \in \mathbb{R}$
 I121: $\alpha, \gamma, \omega \in \mathbb{R}$
 I122: α, γ
 I123: ω
 I124: $\alpha = 0.05, \gamma = 0.4, \omega = 1.3$
 I125: $x(0) \in [0.95, 1.05]$
 I126: $y(0) \in [-0.05, 0.05]$
 I127: $[t_0, t_1] = [0, 100]$
 I128: $A = [-5, 5]$
 I129: $[-5, 5]$
 I130: $N = 1000$
 I131: $\beta = 10^{-9}$
 I132: $s_N^* = 67$
 I133: $\epsilon = 0.2509$
 I134: $\mathbb{R}^2$
 I135: $m = 2$
 I136: $\gamma = 0.25$
 I137: $s_N^* = 19$
 I138: $\epsilon = 0.1182$
 I139: h
 I140: x, h, θ
 I141: u_1
 I142: u_2
 I143: $g = 9.81, K = 0.89/1.4, d_0 = 70, d_1 = 17, n_0 = 55$
 I144: $u_1(t) = u_1$
 I145: $u_2(t) = u_2 \forall t \in [t_0, t_1]$
 I146: $u_1 \in [-1.5 + g/K, 1.5 + g/K], u_2 \in [-\pi/4, \pi/4]$
 I147: $[t_0, t_1] = [0, 5]$
 I148: $A = [-100, 100]^6$
 I149: $\epsilon = 0.1125$
 I150: $m = 3$
 I151: $s_N^* = 22$
 I152: $\epsilon = 0.1253$
 I153: Vol
 I154: $s_N^* \geq d$
 I155: i
 I156: θ_i
 I157: $\mathbb{R}^D$
 I158: μ_i
 I159: σ_i

I160: Φ
I161: ϕ
I162: x, y
I163: $\sin$
I164: $\cos$
I165: $d = 400$
I166: $d = 100$
I167: $s_N^* = 20$